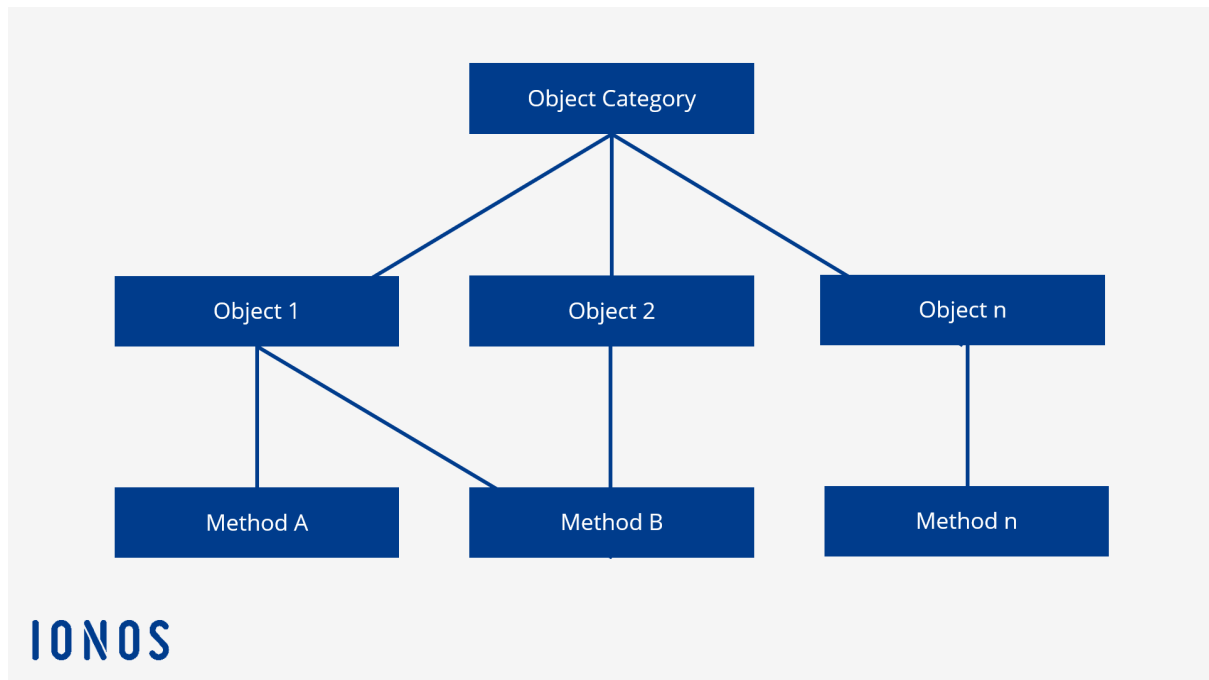


A database is a collection of related data stored on a computer to serve the information retrieval needs of many users and applications.



Database Management System (DBMS): It is software that acts as an intermediary, providing tools to define, manipulate, and control access to data.

Data model Classify data organization methods according to hierarchical, network, and relational models.

Entity: Concrete or abstract objects in the real world that need to be managed are grouped into entity sets.

In diagrams, entities are usually represented by rectangles.

SINHVIEN

KHACHHANG

Attribute Attributes are characteristics that describe entities.

For example:

- Student ID: Student ID, Student Name, Date of Birth

Classify:

- Single-valued attribute
- Multi-valued attribute
- Derived properties

Attributes are often denoted as



The connection: shows the relationship between entities.

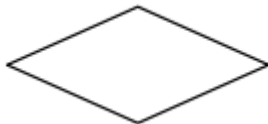
Types:

- 1-1: One to one
- 1-N: One – many
- N-N: Many – many

For example:

- Students – Class $\rightarrow$ 1-N
- Students – Subjects $\rightarrow$ N-N

symbol:



E-R Scheme: This is a diagram used for designing data at the conceptual level.

Ingredient:

- Entity $\rightarrow$ rectangle
- Property $\rightarrow$ ellipse
- Relationship $\rightarrow$ rhombus

Purpose:

- Data modeling before building the database.

Table structureData is represented in the form of relations (tables), where each row is a record and each column is an attribute.

Student ID	TenSV	Lop
001	An	IT

The concept of a lock.:

- **Primary Key:** The minimum set of attributes used to uniquely identify each row in the table.
- **Foreign Key**Used to establish a logical link between different tables.



Difference Between Primary Key and Foreign Key

Primary Key	Foreign Key
Primary key uniquely identify a record in the table.	Foreign key is a field in the table that is primary key in another table.
Primary Key can't accept null values.	Foreign key can accept multiple null value.
By default, Primary key is clustered index and data in the database table is physically organized in the sequence of clustered index.	Foreign key do not automatically create an index, clustered or non-clustered. You can manually create an index on foreign key.
We can have only one Primary key in a table.	We can have more than one foreign key in a table.
The primary key of a particular table is the attribute which uniquely identifies every record and does not contain any null value.	The foreign key of a particular table is simply the primary key of some other table which is used as a reference key in the second table.
A primary key attribute in a table can never contain a null value.	A foreign key attribute may have null values as well.
Not more than one primary key is permitted in a table.	A table can have one or more than one foreign key for referential purposes.
Duplicity is strictly prohibited in the primary key; there cannot be any duplicate values.	Duplicity is permitted in the foreign key attribute, hence duplicate values are permitted.

Convert the diagram: Provides rules for converting from E-R diagrams to corresponding relational tables.

Data Definition Data (DDL) Commands such as **CREATE**, **ALTER**, **DROP** Used to build and modify the structure of tables.

- **CREATE**: create table
- **ALTER**: edit table
- **DROP**: delete table

Data Manipulation (DML) Commands **INSERT**, **UPDATE**, **DELETE** Used to manage the data content within a table.

- **INSERT**: Add data
- **UPDATE**: Corrected data
- **DELETE**: delete data

Data Query Management (DQL) Use the command **SELECT** Combined with conditional clauses, grouping, and sorting, this method extracts information as required.

- **SELECT**: query data

```
SELECT * FROM SinhVien WHERE Lop = 'CNTT';
```

Data redundancy Analyze the negative consequences of poor design, including wasted memory and errors in updating information.

This occurs when data is repeated multiple times.

Harmful effects:

- Uses a lot of memory.
- Easily prone to errors during updates.
- Difficult to maintain.

Functional dependency: It is the relationship between attributes.

Symbol:

- $A \rightarrow B$ (A determines B)

For example:

- $MaSV \rightarrow TenSV$

Standard forms:

1NF (First Normal Form)

- No repeating properties
- Each cell contains the atomic value.

2NF

- Already in 1NF
- No component dependencies

3NF

- No transitive dependency

BCNF

- Stronger than 3NF
- All dependencies originate from keywords.

Target:

- Reduce excess
- Increase consistency

Integrity constraints Business rules mandate that data must comply with regulations to ensure accuracy.

Types:

- Entity Integrity (non-null primary key)
- Referential Integrity (valid foreign key)
- Domain Integrity (valid data type)

Security: Access control and authorization mechanisms to protect the database from unauthorized access.

Include:

- Authentication
- Authorization
- Data encryption

For example:

- Only administrators can delete data.

Transaction management Ensure that all operation groups are executed completely (ACID principle), keeping the system in a consistent state.

- **Atomicity** Either do it all, or do nothing at all.
- **Consistency** The data is always correct.
- **Isolation** The transactions do not affect each other.
- **Durability** Data is not lost after committing.